Monte Carlo Population Synthesis of Galactic Supernova Remnants: Selection Effects and the Missing-Remnant Problem

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The Milky Way contains only a few hundred confirmed supernova remnants (SNRs), whereas independent population arguments indicate that the underlying population may be substantially larger. This discrepancy, commonly referred to as the Galactic "missing-SNR problem", arises because the observed census is affected by both the physical evolution of remnants and the selection properties of radio surveys.

We investigate this problem using a Monte Carlo population-synthesis model in which a synthetic Galactic SNR population is generated, evolved dynamically, assigned radio emission, and subjected to an observational selection function. The reference population is generated from a supernova rate $R_{\rm SN} = 0.02$ yr^{-1} over a reference interval of 10^5 yr. The number of remnants in each realisation is drawn from a Poisson distribution with mean 2000, while remnant ages are sampled uniformly. The population is divided into 80% core-collapse and 20% Type-Ia events. Galactocentric radii follow the pulsar-motivated distribution of Lorimer et al., and an exponential vertical scale height of 80 pc is adopted. Ambient densities are drawn from a log-normal distribution.

The forward-shock evolution is calculated using the analytic Truelove–McKee framework. The 1.4-GHz synchrotron luminosity is calculated with the semi-analytic prescription of Sarbadhicary et al., using the corrected constants and implementation associated with their published erratum. The resulting luminosities are converted to flux densities and to an approximate beam-averaged surface brightness. An illustrative radio selection function requires a surface brightness exceeding 5 mJy beam^{-1} and an angular diameter between 1 and 15 arcmin. These thresholds are used as a controlled reference model and are not identified with a particular historical survey.

Across 500 independent Monte Carlo realisations, the reference model produces a mean of 480.1 ± 14.3 detectable remnants. The reduction from the mean physical population of approximately 2000 objects to the simulated observable population demonstrates the substantial effect of the adopted surface-brightness and angular-scale selection criteria. The undetected population is dominated by evolved remnants, particularly late Sedov–Taylor objects in relatively low-density environments, for which expansion produces large angular sizes and reduced radio surface brightness.

The simulation does not provide a unique estimate of the total Galactic SNR population because its selection function is deliberately idealised and does not reproduce the heterogeneous surveys from which the observational catalogue has been assembled. Nevertheless, the result demonstrates quantitatively that observational selection can substantially alter the apparent Galactic SNR census. The continuing discovery of SNR candidates in sensitive surveys such as the SARAO MeerKAT Galactic Plane Survey provides an empirical context for such forward population modelling.

I. INTRODUCTION

Supernova remnants are important tracers of stellar death, interstellar-medium (ISM) evolution, magnetic-field amplification, and cosmic-ray acceleration. They also provide a time-integrated record of supernova activity over periods much longer than the duration of an individual supernova explosion. Establishing the size and properties of the Galactic SNR population is therefore relevant to several areas of Galactic astrophysics.

The observational census remains incomplete. The most recent published catalogue maintained by Green contains 310 Galactic SNRs [1]. The catalogue is an important observational resource, but its construction is necessarily affected by the sensitivity, angular resolution, spatial-frequency coverage, background emission, and classification properties of the observations used to identify remnants.

Independent population studies have indicated that the intrinsic Galactic population may be considerably larger than the confirmed catalogue. Ranasinghe and Leahy estimated a total Galactic population of approximately 3500 remnants using one of their methods, while alternative models gave values of roughly 2400–5600 objects [2]. These values are model-dependent estimates rather than direct measurements of a known total population. A subsequent statistical analysis likewise demonstrated that selection effects are important in interpreting the observed Galactic radio-SNR sample [3].

Recent radio observations provide direct evidence that the census continues to change. The EMU/POSSUM Galactic pilot field produced 21 SNR candidates and illustrated the effects of small angular size, low surface brightness, background emission, and source confusion on SNR identification [4]. The SARAO MeerKAT Galactic Plane Survey (SMGPS) provides approximately 8″ angu-

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lar resolution and broadband rms sensitivities of about 10–20 μJy beam^{-1} at 1.3 GHz over almost half of the Galactic plane [5]. Anderson et al. subsequently identified 237 new SNR candidates in the SMGPS data. These objects are candidates rather than 237 newly confirmed remnants; confirmation requires additional evidence, such as radio polarization or spectral-index measurements [6].

These observations motivate treating the missing-SNR problem as a forward modelling problem rather than as a simple comparison between the observed catalogue size and a theoretical population estimate. Let $\mathbf{\Theta}$ denote the physical properties of an individual remnant and $\mathbf{\Lambda}$ the parameters describing the underlying population. The population distribution can be written as

$$p(\mathbf{\Theta} \mid \mathbf{\Lambda}), \tag{1}$$

while an observational survey is represented by a selection function $\mathcal{S}(\mathbf{\Theta})$. The expected detected population is then

$$N_{\rm det} = N_{\rm phys} \int \mathcal{S}(\mathbf{\Theta}) p(\mathbf{\Theta} \mid \mathbf{\Lambda})\, {\rm d}\mathbf{\Theta}. \tag{2}$$

The integral in Eq. (2) represents the completeness of the adopted model and selection function.

The commonly used estimate

$$N_{\rm phys} \simeq R_{\rm SN} t_{\rm max} \tag{3}$$

is therefore a population normalisation rather than a prediction of the number of catalogued objects. The visibility of an individual SNR depends on its evolutionary state, environment, distance, angular size, radio emission, and the response of the observing system.

The purpose of this work is to quantify this convolution with a Monte Carlo population-synthesis calculation. The simulation generates a synthetic Galactic population, evolves its remnants using analytic shock-dynamics prescriptions, calculates their radio emission, and applies an explicit observational selection function.

The analysis has two related objectives. First, it determines the observable population produced by a specified reference model. Second, it examines the physical properties of remnants rejected by the reference selection criteria, thereby identifying the evolutionary and observational effects responsible for the difference between the underlying and recovered populations.

The emphasis is on forward modelling. The calculated detection fraction is the completeness of the specified synthetic population under the adopted reference selection function; it is not interpreted as the completeness of the heterogeneous historical Galactic SNR catalogue.

II. THEORETICAL FRAMEWORK

A. Population normalisation

As a reference model we adopt

$$R_{\rm SN} = 0.02 \ {\rm yr}^{-1}, \tag{4}$$

corresponding to two supernovae per century. This value is used as a model parameter rather than as a measurement of the Galactic supernova rate. For comparison, Rozwadowska et al. obtained a core-collapse supernova rate of

$$R_{\rm CC} = 1.63 \pm 0.46 \ (100 \ {\rm yr})^{-1}, \tag{5}$$

illustrating the uncertainty associated with estimates of the Galactic supernova rate [7].

For a reference interval of

$$t_{\rm max} = 10^5 \ {\rm yr}, \tag{6}$$

the expected number of remnants is

$$\langle N_{\rm phys} \rangle = R_{\rm SN} t_{\rm max} = 2000. \tag{7}$$

The final Monte Carlo implementation samples the number of remnants in each realisation from

$$N_{\rm phys} \sim {\rm Poisson}(R_{\rm SN} t_{\rm max}), \tag{8}$$

so that the physical population fluctuates statistically between realisations. The corresponding Poisson standard deviation of the reference population is

$$\sigma_N = \sqrt{2000} \simeq 44.7. \tag{9}$$

For a stationary supernova rate, conditional on the number of remnants, the age distribution is uniform:

$$p(t) = \frac{1}{t_{\rm max}}, \qquad 0 < t < t_{\rm max}. \tag{10}$$

Thus, the population model describes remnants with ages uniformly distributed between zero and 10^5 yr. The upper age is a modelling boundary for the synthetic population rather than a universal physical lifetime for radio emission.

B. Progenitor classes and spatial distribution

The reference population is divided into core-collapse and Type-Ia events according to

$$f_{\rm CC} = 0.8, \qquad f_{\rm Ia} = 0.2. \tag{11}$$

These fractions are adopted as reference parameters and are not inferred from the simulation.

The Galactocentric radial distribution is based on the pulsar-motivated prescription of Lorimer et al. [8]:

$$f(R) = A \left(\frac{R}{R_\odot} \right)^B \exp\left[-C \left(\frac{R - R_\odot}{R_\odot} \right) \right], \tag{12}$$

where the simulation adopts

$$R_\odot = 8.5 \ {\rm kpc}, \qquad B = 1.9, \qquad C = 5.1. \tag{13}$$

The radial distribution is sampled using rejection sampling.

The Lorimer distribution is used as a phenomenological proxy for the Galactic distribution of supernova progenitors. It is not assumed to be an exact empirical SNR radial distribution because the observed SNR population itself is subject to selection effects.

The vertical distribution is represented by a two-sided exponential:

$$p(z) = \frac{1}{2h_z} \exp\left(-\frac{|z|}{h_z}\right), \tag{14}$$

with

$$h_z = 80 \text{ pc}. \tag{15}$$

The same scale height is adopted for the entire synthetic population. This is a simplifying approximation that is most naturally associated with the young core-collapse population. Type-Ia progenitors can have broader vertical distributions because of their older stellar ages, and this constitutes an important limitation of the reference model.

For a Galactocentric radius R, azimuth ϕ, and vertical coordinate z, the Cartesian coordinates are

$$x = R\cos\phi, \qquad y = R\sin\phi, \tag{16}$$

and the heliocentric distance is

$$d^2 = (x - R_\odot)^2 + y^2 + z^2. \tag{17}$$

C. Ambient density and explosion properties

The ambient hydrogen number density is sampled from a log-normal distribution. In the final implementation,

$$\ln n_0 \sim \mathcal{N}(-1.33, 0.49^2), \tag{18}$$

where n_0 is expressed in cm^{-3}.

This prescription is phenomenological and is intended to represent the broad range of densities encountered by remnants. It does not attempt to reproduce the full multiphase structure of the Galactic ISM. In particular, the simulation treats the position and ambient density as independent variables, whereas real core-collapse explosions are correlated with recent star formation and hence with particular ISM environments.

The explosion energy is fixed at the reference value

$$E_0 = 10^{51} \text{ erg}, \qquad E_{51} = 1, \tag{19}$$

for both progenitor classes.

For core-collapse remnants, the ejecta mass is sampled from

$$M_{\rm ej,CC} \sim \mathcal{N}(5.0, 1.5^2)\ M_\odot, \tag{20}$$

whereas Type-Ia remnants are assigned

$$M_{\rm ej,Ia} = 1.4\ M_\odot. \tag{21}$$

These choices provide a simple representation of the two progenitor classes while retaining stochastic variation in the core-collapse ejecta mass.

D. Shock evolution

The dynamical evolution is calculated using the analytic framework of Truelove and McKee [9]. The characteristic scales are

$$R_{\rm ch} = \left(\frac{3M_{\rm ej}}{4\pi\rho_0}\right)^{1/3}, \tag{22}$$

$$v_{\rm ch} = \left(\frac{2E_0}{M_{\rm ej}}\right)^{1/2}, \tag{23}$$

and

$$t_{\rm ch} = \frac{R_{\rm ch}}{v_{\rm ch}}. \tag{24}$$

The corresponding dimensionless transition time depends on the progenitor class. The implementation uses

$$t_{\rm ST} = t_*\, 423 \text{ yr}\, E_{51}^{-1/2} M_{\rm ej}^{5/6} n_0^{-1/3}, \tag{25}$$

where

$$t_* = \begin{cases} 0.481, & \text{Ia}, \\ 0.424, & \text{CC}. \end{cases} \tag{26}$$

The ejecta-dominated phase is represented by the appropriate Truelove–McKee dimensionless solution for each progenitor class. Once the remnant passes the transition to the Sedov–Taylor regime, the radius follows the characteristic scaling

$$R_{\rm ST} \propto \left(\frac{E_0}{\rho_0}\right)^{1/5} t^{2/5}. \tag{27}$$

For a monatomic gas, the corresponding dimensional expression may be written approximately as

$$R_{\rm ST} \simeq 1.15 \left(\frac{E_0}{\rho_0}\right)^{1/5} t^{2/5}, \tag{28}$$

where the numerical coefficient depends on the adopted normalisation and adiabatic index.

The Sedov–Taylor relation demonstrates the central dynamical property relevant to the missing-remnant problem: the shock continues to expand while its velocity decreases. Consequently, evolved remnants can occupy increasingly large angular areas while their radio surface brightness becomes progressively more difficult to detect.

The calculation is based on the non-radiative Truelove–McKee evolution used in the final simulation module. Radiative cooling is not separately imposed as an instantaneous radio-emission cutoff. Consequently, the interpretation of very old remnants should be regarded as conditional on the adopted non-radiative dynamical and radio-emission prescriptions.

E. Radio synchrotron emission

The radio emission from Galactic SNRs is predominantly non-thermal synchrotron radiation. A useful dimensional scaling for the shock-powered radio luminosity is

$$L_\nu \propto \rho_0 R_s^2 v_s^3, \tag{29}$$

but this relation alone does not determine an absolute luminosity because the electron acceleration efficiency and magnetic-field amplification must also be specified.

The present calculation uses the semi-analytic prescription of Sarbadhicary et al. [10]. The implementation incorporates the corrections specified in their 2019 erratum [11]. In particular, the corrected upstream magnetic-field prescription is

$$B_0 = 9\ \mu\mathrm{G}\left[\frac{\rho_0}{1.4(1.67\times10^{-24}\,\mathrm{g\,cm^{-3}})}\right]^{0.47}. \tag{30}$$

The accelerated electron population is represented by

$$N(E) = N_0 E^{-p}, \tag{31}$$

with

$$p = 2.5 \tag{32}$$

in the numerical implementation. The electron acceleration efficiency is

$$\epsilon_e = 4.2\times10^{-3}. \tag{33}$$

The electron normalisation is

$$N_0 = (p-2)\epsilon_e \rho_0 v_s^2 E_m^{p-2}, \qquad E_m = m_e c^2. \tag{34}$$

The Alfvén speed and Alfvén Mach number are

$$v_A = \frac{B_0}{\sqrt{4\pi\rho_0}}, \tag{35}$$

and

$$M_A = \frac{v_s}{v_A}. \tag{36}$$

The amplified magnetic field is then calculated using the Mach-number-dependent prescription implemented by Sarbadhicary et al. The final numerical module uses the corrected constants

$$c_1 = 6.27\times10^{18}, \qquad c_5 = 9.68\times10^{-24}, \qquad c_6 = 8.1\times10^{-41}. \tag{37}$$

For an optically thin shell, the synchrotron luminosity can be written in the form

$$L_\nu = 4\pi^2 R_s^2 \left(\frac{c_5}{c_6}\right) B^{-1/2}\left[1-\exp\left\{-\left(\frac{\nu}{\nu_1}\right)^{-(p+4)/2}\right\}\right] \times\left(\frac{\nu}{2c_1}\right)^{5/2}, \tag{38}$$

where

$$\nu_1 = 2c_1(sc_6N_0)^{2/(p+4)}B^{(p+2)/(p+4)}, \tag{39}$$

and the effective emitting-shell thickness is

$$s = \epsilon_f\frac{4R_s}{3}, \qquad \epsilon_f = 0.38. \tag{40}$$

The reference observing frequency is

$$\nu = 1.4\ \mathrm{GHz}. \tag{41}$$

The numerical implementation uses the corrected Sarbadhicary light-curve module throughout the Monte Carlo calculation. This is preferable to using the uncorrected compact expressions of the original publication because the 2019 erratum identified corrections to the magnetic-field normalisation and other constants [11].

III. MATERIALS AND METHODS

A. Observable quantities

For a source at heliocentric distance d, the flux density is

$$S_\nu = \frac{L_\nu}{4\pi d^2}. \tag{42}$$

The angular diameter is approximated by

$$\theta \simeq \frac{2R_s}{d}, \tag{43}$$

with the resulting angle converted to arcminutes. The exact small-angle conversion is appropriate for the angular sizes considered in the simulation.

For an extended source, the integrated flux is distributed over multiple synthesized beams. The number of beams is approximated by

$$N_{\mathrm{beam}} = \max\left[1, \left(\frac{\theta}{\theta_{\mathrm{beam}}}\right)^2\right], \tag{44}$$

where

$$\theta_{\mathrm{beam}} = 1\ \mathrm{arcmin} \tag{45}$$

is the reference beam size.

The characteristic flux density per beam is therefore

$$S_{\nu,\mathrm{beam}} = \frac{S_\nu}{N_{\mathrm{beam}}}. \tag{46}$$

This quantity is used as an approximate mean surface-brightness measure. It is not the peak brightness of a resolved shell. A detailed survey simulation would require a spatially resolved shell-emissivity profile, beam convolution, and an explicit source-detection algorithm.

An important consequence of Eqs. (42)–(46) is that for a sufficiently resolved source, both integrated flux and projected solid angle scale approximately as d^{-2}. Surface brightness is therefore much less sensitive to distance than integrated flux. Distance nevertheless affects detectability through angular size, beam dilution, source confusion, and interferometric spatial-frequency response.

B. Reference selection function

The simulation applies three observational criteria:

$$S_{\nu,\mathrm{beam}} > 5\ \mathrm{mJy\,beam^{-1}}, \tag{47}$$

$$\theta > 1\ \mathrm{arcmin}, \tag{48}$$

and

$$\theta < 15\ \mathrm{arcmin}. \tag{49}$$

These conditions constitute an illustrative reference selection function. They are not identified with a particular historical VLA survey. The 5 $\mathrm{mJy\,beam^{-1}}$ threshold represents a controlled reference surface-brightness limit, while the angular criteria provide idealised small- and large-scale filtering.

The corresponding selection function can be written

$$\mathcal{S}_{\mathrm{ref}} = H(S_{\nu,\mathrm{beam}} - S_{\mathrm{lim}})H(\theta - \theta_{\mathrm{min}})H(\theta_{\mathrm{max}} - \theta), \tag{50}$$

where H denotes the Heaviside function.

The upper angular limit is not treated as a universal interferometric maximum recoverable scale. The maximum recoverable angular scale depends on observing frequency, minimum baseline, weighting, and imaging strategy. Likewise, the 1-arcmin lower limit is not a fundamental physical or instrumental limit. These values are used solely to isolate the effects of surface-brightness and angular-scale selection in the reference experiment.

C. Monte Carlo implementation

The final simulation consists of 500 independent Monte Carlo realisations. For each realisation, the following sequence is applied:

1. draw the number of SNRs from the Poisson distribution in Eq. (8);
2. assign each remnant to the core-collapse or Type-Ia population with probabilities 0.8 and 0.2;
3. sample the remnant age uniformly between 0 and 10^5 yr;
4. sample its Galactocentric radius using the Lorimer et al. distribution;
5. sample the azimuth uniformly between 0 and 2π;
6. sample the vertical position from the two-sided exponential distribution with scale height 80 pc;
7. calculate the heliocentric distance;
8. sample the ambient density from the log-normal distribution;
9. assign the explosion energy and ejecta mass;
10. calculate the shock radius and velocity using the Truelove–McKee-based module;
11. calculate the 1.4-GHz synchrotron luminosity using the corrected Sarbadhicary prescription;
12. calculate the integrated flux density and angular diameter;
13. calculate the characteristic surface brightness per beam;
14. apply the three observational selection criteria; and
15. record the number of detected remnants for the realisation.

The Monte Carlo ensemble is therefore characterised by

$$N_{\mathrm{MC}} = 500. \tag{51}$$

The mean detected population is calculated as

$$\overline{N}_{\mathrm{det}} = \frac{1}{N_{\mathrm{MC}}} \sum_{k=1}^{N_{\mathrm{MC}}} N_{\mathrm{det},k}, \tag{52}$$

with ensemble standard deviation

$$\sigma_{\mathrm{MC}} = \left[\frac{1}{N_{\mathrm{MC}} - 1} \sum_{k=1}^{N_{\mathrm{MC}}} \left(N_{\mathrm{det},k} - \overline{N}_{\mathrm{det}} \right)^2 \right]^{1/2}. \tag{53}$$

The final simulation records the total number of survivors from each realisation. The underlying object-level quantities used to determine detectability are generated within the simulation and include age, density, shock radius, shock velocity, distance, angular diameter, luminosity and flux density.

IV. RESULTS AND DISCUSSION

A. Reference population

The reference supernova rate and time interval give an expected physical population of

$$\langle N_{\mathrm{phys}} \rangle = 0.02 \times 10^5 = 2000. \tag{54}$$

Because the number of remnants is drawn from a Poisson distribution, individual realisations fluctuate around this expectation. The corresponding intrinsic population standard deviation is

$$\sigma_{N_{\mathrm{phys}}} = \sqrt{2000} \simeq 44.7. \tag{55}$$

The observational selection function substantially reduces the recovered population. Across the 500 independent Monte Carlo realisations, the simulation gives

$$\overline{N}_{\mathrm{det}} = 480.1 \pm 14.3 \tag{56}$$

TABLE I. Reference Monte Carlo parameters and final results.

Quantity	**Value**
Supernova rate	0.02 yr^{-1}
Maximum age	10^5 yr
Expected physical population	2000
Monte Carlo realisations	500
CC fraction	0.80
Type-Ia fraction	0.20
Vertical scale height	80 pc
Reference surface-brightness limit	5 $\mathrm{mJy\,beam}^{-1}$
Minimum angular diameter	1 arcmin
Maximum angular diameter	15 arcmin
Mean detected population	480.1
Monte Carlo standard deviation	14.3
Reference-model detection fraction	0.240

where the uncertainty denotes the standard deviation of the detected counts across the Monte Carlo ensemble.

The corresponding mean detection fraction of the synthetic reference population is

$$f_{\mathrm{det}} = \frac{\overline{N}_{\mathrm{det}}}{\langle N_{\mathrm{phys}} \rangle} = \frac{480.1}{2000} \simeq 0.240. \tag{57}$$

Thus, under the specific physical and observational assumptions of the reference simulation, approximately 24% of the synthetic remnants satisfy all three detection criteria.

This fraction should not be interpreted as the completeness of the actual Galactic SNR catalogue. The reference selection function is intentionally idealised and does not represent the heterogeneous collection of surveys, frequencies, sensitivities, angular resolutions, sky coverage, background conditions, and classification procedures that produced the observed catalogue.

B. Interpretation of the observational selection

The reduction from approximately 2000 physical remnants to a mean of 480.1 detectable objects arises from the combined action of the surface- brightness and angular-size criteria. The simulation therefore demonstrates that a substantial fraction of an underlying Galactic SNR population can fail an idealised radio selection function even when the remnants are physically present.

The surface-brightness criterion is particularly important for evolved remnants. During the Sedov–Taylor stage the shock radius increases as approximately $t^{2/5}$ while the shock velocity decreases with time. Consequently, radio emission is distributed over an increasingly large projected area. The characteristic flux density per synthesized beam can therefore decrease even when the remnant remains physically extended and contains an energetic shock.

The angular-size criterion acts in two directions. Remnants smaller than 1 arcmin are excluded by the adopted lower limit, whereas sufficiently evolved nearby or large remnants can exceed 15 arcmin and are rejected by the adopted large-scale criterion. The latter condition provides a simple representation of the loss of sensitivity to sufficiently extended emission that can occur in interferometric observations.

These effects are not mutually exclusive. An evolved remnant can be both large and faint, so the observed incompleteness is naturally a multidimensional selection effect rather than the result of a single threshold.

C. Characteristics of the undetected population

The properties of the remnants rejected by the reference selection show that the missing synthetic population is dominated by evolved objects. In particular, late Sedov–Taylor remnants in relatively low-density environments form an important component of the rejected population.

The physical reason is straightforward. As the shock expands, its angular diameter increases while its velocity decreases. The associated radio emission is spread over a larger projected area, reducing the characteristic surface brightness. At the same time, sufficiently large remnants exceed the adopted 15-arcmin angular-scale limit.

The simulation therefore links the missing-remnant problem directly to SNR evolution: the remnants most difficult to recover with the reference selection are not necessarily physically absent, but can occupy a region of parameter space in which large angular extent and low surface brightness combine to reduce detectability.

The result is consistent with the broader observational picture in which large, diffuse Galactic radio structures can be difficult to recognise against complex Galactic emission and can be affected by interferometric spatial filtering. It does not, however, imply that every undetected Galactic SNR must have the same evolutionary properties.

D. Role of the Galactic environment

The density distribution introduces an additional physical selection effect. For a fixed explosion energy, the ambient density affects both the shock dynamics and the radio emission. Higher densities increase the amount of material interacting with the shock, but also modify the expansion rate and the subsequent evolution of the remnant. Lower-density environments allow more rapid expansion and can produce large, diffuse remnants whose radio surface brightness becomes difficult to detect.

The reference model deliberately treats the ambient density as an independent random variable. Real Galactic remnants occupy a structured multiphase ISM, and core-collapse explosions in particular are associated with regions of recent massive-star formation. The simulated density distribution should therefore be regarded as a

phenomenological representation rather than a complete model of the Galactic ISM.

Similarly, the common 80-pc scale height adopted for both progenitor classes is a simplification. Type-Ia progenitors can occupy larger vertical distances from the Galactic plane. Such environmental differences can alter both the ambient density and the radio detectability of the resulting remnants.

E. Comparison with the observed Galactic catalogue

The Green catalogue contains 310 confirmed Galactic SNRs [1]. The reference simulation produces a larger observable synthetic population, with a mean of 480.1 objects. This numerical comparison is useful for illustrating the scale of the selection problem, but it should not be interpreted as a direct prediction of the number of objects that a historical catalogue should contain.

The reason is that the reference selection function is not the selection function of the Green catalogue. The observed catalogue contains remnants identified using observations made at different frequencies and with different sensitivities, angular resolutions, sky coverage, interferometric configurations, and background conditions.

A more appropriate representation is

$$N_{\rm cat} = \int \lambda(\boldsymbol{x} \mid \boldsymbol{\Lambda}) \mathcal{S}(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x}, \tag{58}$$

where λ describes the intrinsic SNR population and $\mathcal{S}$ is the actual survey-and-classification selection function.

Consequently, the difference between 480.1 simulated detections and 310 confirmed catalogue objects should not be interpreted as evidence that the simulation overpredicts the physical population by a particular factor. Rather, it illustrates the sensitivity of catalogue recovery to the assumed selection function.

F. Comparison with modern radio surveys

The SMGPS provides an important observational context for the present model. It operates at approximately 1.3 GHz with $8''$ angular resolution and broadband rms sensitivities of approximately 10–20 μJy beam^{-1} over almost half of the Galactic plane [5].

Anderson et al. identified 237 new SNR candidates in the SMGPS data, together with the expected radio morphology of 201 previously known SNRs and 130 previously identified candidates in the survey area [6]. The 237 new objects are candidates rather than confirmed SNRs, and additional evidence such as radio polarization or spectral-index measurements is required for confirmation.

The large number of candidates identified in the SMGPS illustrates why surface brightness, angular resolution, confusion and spatial-frequency coverage must be included when interpreting the Galactic SNR census. The present simulation captures only a simplified subset of these effects.

In particular, the reference model does not reproduce the actual SMGPS selection function. It does not include the detailed survey footprint, spatially varying noise, synthesized-beam response, *uv* coverage, diffuse Galactic background, or the probability of classifying a detected radio structure as an SNR. The comparison with SMGPS is therefore qualitative rather than a direct survey-to-simulation prediction.

G. Limitations of the reference model

Several simplifications determine the quantitative value of the reported detection fraction.

First, the radial distribution is based on a pulsar population model rather than a dedicated empirical SNR progenitor distribution. Second, a common 80-pc vertical scale height is used for both core-collapse and Type-Ia events. Third, the ambient density is sampled independently of Galactic position and progenitor type.

Fourth, the shock evolution is represented analytically using the Truelove–McKee framework rather than through multidimensional hydrodynamic simulations. Fifth, the radio emission model is semi-analytic and depends on assumptions concerning electron acceleration and magnetic-field amplification. Finally, the observational selection function represents a controlled reference experiment rather than an actual historical survey.

These limitations do not invalidate the Monte Carlo experiment. Instead, they define the interpretation of its numerical result: 480.1 ± 14.3 is the predicted recovered population for the specified synthetic Galaxy, physical prescriptions, and reference selection function.

H. Implications for population inference

The simulation demonstrates why the Galactic missing-SNR problem cannot be reduced to the relation

$$N_{\rm phys} = R_{\rm SN} t_{\rm max} \tag{59}$$

alone. The observed number depends on the convolution of the intrinsic population with evolutionary physics and observational selection.

A statistically rigorous Galactic population inference would therefore require a survey-specific selection function and a likelihood describing the observed distribution of SNR properties. In a Poisson point-process formulation, one may write schematically

$$\ln \mathcal{L} = -N_{\rm exp}(\boldsymbol{\Lambda}) + \sum_i \ln \lambda(\boldsymbol{x}_i \mid \boldsymbol{\Lambda}), \tag{60}$$

where

$$N_{\rm exp}(\boldsymbol{\Lambda}) = \int \lambda(\boldsymbol{x} \mid \boldsymbol{\Lambda}) \mathcal{S}(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x}. \tag{61}$$

Such a framework would allow the supernova rate, spatial distribution, ambient-density distribution, radio microphysics and survey selection parameters to be constrained jointly. The present Monte Carlo calculation provides a simpler forward-modelling demonstration of the same underlying principle.

V. CONCLUSIONS

We have investigated the Galactic missing-SNR problem using a Monte Carlo population-synthesis model that combines supernova-population sampling, Galactic spatial distributions, analytic shock evolution, synchrotron radio emission and an explicit observational selection function.

The reference model adopts a supernova rate of $0.02\ \mathrm{yr}^{-1}$ and a maximum synthetic age of 10^5 yr, giving an expected physical population of 2000 remnants. The number of remnants in each realisation is sampled from a Poisson distribution, while ages are distributed uniformly. The population is divided into 80% core-collapse and 20% Type-Ia events, with a Lorimer et al. radial distribution and an 80-pc exponential vertical scale height.

The shock evolution is calculated using the Truelove–McKee framework, while the radio luminosity is calculated using the corrected semi-analytic Sarbadhicary et al. prescription. The resulting flux density and angular size are passed through a reference selection function requiring $S_{\nu,\mathrm{beam}} > 5\ \mathrm{mJy\,beam}^{-1}$ and an angular diameter between 1 and 15 arcmin.

For 500 independent Monte Carlo realisations, the reference model produces

$$\overline{N}_{\mathrm{det}} = 480.1 \pm 14.3. \tag{62}$$

Relative to the expected physical population of 2000 remnants, this corresponds to a detection fraction of approximately 0.240 under the reference selection function.

The simulation therefore demonstrates quantitatively that observational selection can strongly reduce the number of remnants recovered from an underlying Galactic population. The rejected synthetic remnants are predominantly evolved objects, particularly late Sedov–Taylor remnants in relatively low-density environments. Their large physical and angular sizes, together with reduced radio surface brightness, make them particularly susceptible to the adopted observational cuts.

The result should not be interpreted as a direct estimate of the total number of Galactic SNRs. The reference selection function is not the heterogeneous selection function of the Green catalogue, and the physical model contains simplifications in the spatial distribution, ISM density, progenitor populations, shock evolution and radio microphysics.

Nevertheless, the calculation provides a quantitative demonstration of the central role of observational selection in the missing-remnant problem. The continued discovery of SNR candidates by sensitive surveys such as SMGPS, including the 237 candidates reported by Anderson et al., provides an observational context in which such selection effects are particularly relevant.

A quantitative determination of the intrinsic Galactic SNR population would require survey-calibrated forward modelling incorporating the actual survey footprint, sensitivity variations, angular response, interferometric spatial-frequency coverage, diffuse Galactic emission, source confusion and classification probability. Until such effects are incorporated, the scientifically defensible conclusion from the present model is that observational selection can account for a substantial reduction between an underlying synthetic SNR population and its recovered radio sample, while the intrinsic Galactic population itself cannot be uniquely determined from the reference selection model alone.

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CODE AND DATA AVAILABILITY

The Monte Carlo implementation used for the reported calculation is based on the Python simulation and the associated Sarbadhicary radio-emission module documented in the project materials. The parameter values and numerical procedure required to reproduce the reference calculation are specified in the present paper and its supplementary code listing.

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